

REI603M - Final Project

Spring 2026

1. Overview

This is the culmination of your semester-long project. You will present a working AI-powered application to the class, demonstrate it live, and submit a written report covering how you built it and what you learned.

Work in the **same groups as throughout the semester**. The final project is worth **30% of your final course grade**.

2. Submission Requirements

- **Deliverables:** Submit a .zip file on Canvas containing:
 1. Presentation slides (PDF)
 2. Written report: REI603M_FinalProject_usernames.md where *usernames* is an underscore-separated list of your university usernames
- **Git repository:** Already shared with haffi112 on GitHub. Must include a README.md with setup/run instructions and a .env.example showing required environment variables (without real keys)
- **Deadline:** Before your assigned presentation slot in the last two weeks of the semester. Exact dates will be decided in class.
- **Presentations:** 25 minutes per group, including a live demo

Hard rules:

- If slides are not submitted before the presentation, the final project is **marked as failed**
- If the written report is not submitted, the maximum grade is **5.0**
- The demo must be **live**. The application must be running during the presentation

3. Written Report

Your report must follow the same 8-section structure described below. Write it as a Markdown document. Be concise and specific. There is no strict page limit, but aim for substance over length.

4. Product Dimensions

Your presentation and report must cover these 8 sections. Each section has specific requirements listed below.

4.1. Problem & Users

- Define the problem you are solving in concrete terms
- Identify your target users: who they are, what they currently do without your product
- Explain why an AI/LLM-based solution is appropriate for this problem
- Describe how your understanding of the problem evolved since Assignment 1

4.2. Technical Architecture

- Include a system diagram showing all components: UI, backend, LLM APIs, database, external services, and how they connect
- Explain your key technology choices and the trade-offs behind them
- Describe how data flows through the system from user input to output
- Describe how prompts are managed: where they are stored, how they are versioned, how you iterate on them

4.3. LLM Integration & Prompt Strategy

- Which models do you use and why
- What prompting techniques worked for your use case (few-shot, structured outputs, tool use, RAG, agents, etc.)
- How your prompts evolved from Assignment 3 through the final version
- Any advanced patterns you implemented: multi-turn context management, function calling, multi-model routing, or others

4.4. Evaluation & Quality Assurance

- Describe your evaluation methodology and dataset, building on what you developed in Assignment 3
- Present quantitative and/or qualitative results on your final system
- Analyze where the system fails and how you handle those failures
- Describe any guardrails, safety measures, or output validation you implemented

4.5. Deployment & Monitoring

- Where and how the application is deployed
- What observability you have in place, building on Assignments 4 and 6
- What metrics you track: latency, cost, token usage, quality signals, or others
- Show a dashboard, trace, or monitoring view from real usage

4.6. Feature Plan Retrospective

In Assignment 6, you listed at least 5 features you planned to complete before the final submission. Revisit that list:

- For each planned feature: did you complete it, partially complete it, or drop it?
- Explain why for any features that were descope or changed
- Describe any features you added that were not on the original A6 list

4.7. User Testing & Interaction Data

You must test your application with real users who are **not** members of your group.

User testing requirements:

- **5 or more external users:** full points
- **Fewer than 5:** partial points, proportional to the number of users tested
- Group members do not count as external users

For this section:

- Describe your testing methodology: what you asked users to do, how you collected feedback
- Present findings: what worked, what confused users, what broke

- Show interaction data you collected and what it revealed about usage patterns
- Explain how user feedback informed improvements to the product

4.8. Lessons Learned & Next Iteration

- What would you do differently if you started over
- What would the next iteration of the AI lifecycle look like for this product: what data would you collect, what would you change, what features would you prioritize
- Your most important technical and non-technical takeaway from the project

5. Presentation Structure

Your **25-minute** presentation should follow this structure. **Slides 8–10 are a live demo** (5–7 minutes).

Slide	Content
1. Title + Team	Project name, team members
2. Problem & Users	What problem you solve, who your users are
3. Technical Architecture	System diagram, technology choices, data flow
4. LLM Integration	Models used, prompting strategy, advanced patterns
5. Evaluation & Quality	Methodology, results, error analysis, guardrails
6. Deployment & Monitoring	Where deployed, observability setup, metrics and dashboards
7. Feature Retrospective	A6 plan vs. reality: what shipped, what was cut, what was added
8–10. LIVE DEMO	Show the primary user flow end-to-end, then show monitoring and observability
11. User Testing	Methodology, findings, interaction data, how feedback shaped the product
12. Lessons Learned	Reflections, next iteration, key takeaways

All group members should participate in the presentation.

6. Grading

Your grade is based on **submitting your work and presenting according to the instructions**.

Deductions: –0.5 for each missing element listed below.

Report & Deliverables

- Missing `.env.example` or secrets committed to the repository
- README lacks setup or run instructions
- Git repository not shared with haffi12

Problem & Users

- No clear problem definition
- Target users not identified
- No justification for an AI/LLM-based approach

Technical Architecture

- No system diagram
- Key technology choices not explained
- No description of data flow
- Prompt management approach not described

LLM Integration & Prompt Strategy

- No explanation of model choice
- No discussion of prompting techniques used
- No evidence of prompt evolution since Assignment 3

Evaluation & Quality Assurance

- No evaluation methodology or dataset
- No quantitative or qualitative results presented
- No error analysis
- No guardrails or output validation discussed

Deployment & Monitoring

- Application not deployed
- No observability beyond the Assignment 4 baseline
- No metrics tracked or shown
- No dashboard, trace, or monitoring view presented

Feature Plan Retrospective

- Assignment 6 feature plan not revisited
- No accounting of which features were completed, descoped, or added

Live Demo

- No live demo
- Demo does not show the primary user flow
- Demo does not show monitoring or observability

User Testing & Interaction Data

- No user testing conducted (significant deduction)
- Fewer than 5 external users tested (partial deduction, proportional)
- No description of testing methodology
- No findings or user feedback presented
- No interaction data shown

Lessons Learned & Next Iteration

- No reflection on what you would do differently
- No discussion of the next lifecycle iteration

For any questions, contact: hafsteinne@hi.is. When asking for help, provide context and explain what you have tried so far.